The Corporate Heist - Solution Documentation

Team name: [TEAM NAME] **Unstop team ID:** [UNSTOP TEAM ID] **Members:** Arush Pratap Singh, [other members]

1. Summary

Team name: Giants Unstop team ID: Members: Arush Pratap Singh, Himesh, Raj Kurele, Shaurya Pratap Singh

"proper" degree, CV without gap) are learnt by the model but then *neutralised* when scoring the Vault. (ii) The insider's five statements are turned into rules: fast-track for sustained public code contributions, a boost for assessment toppers from colleges never hired from before, a retained leg-up for a few old-boys-network colleges, and hard exclusions for title-inflated CVs and candidates who cannot join within two months. (iii) Fabricated profiles and repeated persons are removed before ranking. Of 10,000 Vault rows, 996 are excluded; the top 500 of the remaining are ranked by score. The shortlist contains 116 sustained contributors (90+ contributions), 124 further profiles with moderate contribution credit, 54 assessment toppers from new colleges, and the rest are high-model-score candidates (mean predicted score 70.5, mean assessment 78, mean role-fit 5.0 of 6 core skills).

2. Data cleaning and parsing

All parsing lives in `code/heist_clean.py`. Every column was normalised to one unit; unparseable values become missing (never guessed).

Field	Messy values found	Handling
total_experience	"Fresher", "22 months", "20+ years", ">20", "<1 year", "8+ yrs", "17.6 years", 4.6	years (float); Fresher=0, months/12, "N+"=N+0.5, ">20"=21, "<1"=0.5
technical_assessment	"0.43", "49/100", "99.0 %", 86, "absent", "not taken"	scaled 0-100; fractions x100; absent/not taken = missing + flag
aptitude_score	"5.7/10", "55%", 6.3	0-10 scale (percent/10)
current/expected_ctc	"58.3 LPA", "5.40L", "INR 54.8 lakh", "Rs 38,33,000", "3360000", "7.53 Cr", "$31,230"	annual INR (LPA/lakh x1e5, Cr x1e7, $ x83)
notice_period	"1 month", "60 days", "Immediate joiner", "Available now", "Serving notice - 30 days", "4 months", "180 days"	days (month=30) + serving flag
last_rating, kpi_met, awards	"Outstanding", "3/5", "New joiner - not rated"; Y/Yes/1/TRUE/no/0/FALSE; "-", "1 award", "Yes - Spot Award"	1-5 numeric (text mapped), 0/1 flags
institute (944 spellings)	"IIT-D", "I.I.T. Delhi", "IITB", "INDIAN INSTITUTE OF TECHNOLOGY DELHI", " IIT Delhi ", "NIT Trichy", "BITS, Pilani", "DTU"/"Delhi College of Engineering (DCE)"	case/space/punctuation normalisation + alias table -> ~400 canonical colleges
city, degree	Madras/Chennai, Gurgaon/Gurugram, Calcutta/Kolkata, New Delhi/Delhi NCR; B.E./BE/Btech/Bachelor of Technology	canonical city and degree codes
skills (197 spellings)	"Python3"/"python 3"/"PYTHON", "k8s"/"EKS/GKE", "AB testing"/"split testing"	alias map -> 56 canonical skills, multi-hot
career_path	separators ->, >, |, arrows; "(3.4y)", "[23 mo]", "(1.6 yrs)"	parsed to (title, years); seniority level per title
public_code_ contributions	"~3", "10+", "3 PRs", "3 merged PRs", "not tracked", blank, 60, 408	count (N+ = N+0.5); "not tracked"/blank = 0 credit
name / email / phone	"Dadhich, Ashutosh" vs "Ashutosh Dadhich"; "x+jobs@..." ; "+91-61173-08690" vs "06117308690"	used only to detect repeated persons (section 5)

3. What drives a great hire - findings

- **Fabricated profiles are the biggest trap.** Assessment score rises steadily to ~63 average post-hire score at 77-84, then *collapses* (43 above 84). Cause: rows with assessment >= 85 *and* two stacked positive recruiter notes (625 rows, 3.1%) have mean score 9.1 (median 0), while equally high scorers without the double note average 68.0 (n=835). Every timeline-impossible row (experience longer than time since degree, degree before age 18, age minus experience < 18) falls inside this set.
- **Role fit**: each applied role has 6 core skills (prevalence ~0.6 among its applicants vs ~0.1 for others). Score climbs from 36 (0 core skills) to 64 (all 6), ~+4.8 per core skill; off-role skills add nothing and long skill lists are a fabrication signal.
- **Other genuine drivers** (ridge coefficients, points): KPI met +3.1, each rating point +2.4, award +1.5, technical assessment +0.27/pt (+0.22 more above 70), one positive note +1.5 / negative note -1.5, overtime history -1.0, higher CTC/seniority mildly negative (long-tenure leaders score about 4-5 points below mid-career).
- **Old-panel preferences** (relative points): college tier (IIT/NIT/BITS/IISc/DTU/NSUT ~ +6 to +9 vs mid colleges), metro city (Mumbai/Chennai/Pune/Hyderabad/Delhi ~ +3.5 vs Indore/Patna/Jaipur ~ -5), referral channel +4.9, employer size 5,000+ ~ +4, B.Tech/master's degree ~ +5 vs 12th/BCA/BSc, CV gap of more than 2 years since degree -3 (sharp step at 2 years). Their coefficients on the true drivers are unchanged with/without these terms, so they are additive biases, not confounders.
- **Validation** (Ledger, dev_winners): a model trained on the Archive with all features recovers **112 of the 150** Ledger top-5% in its top 150 (repeat persons collapsed); 5-fold CV R^2 on genuine Archive rows = 0.42. The preference-neutralised model recovers 79/150 on the Ledger - expected, because the Ledger itself was scored by the biased panel; we use it for the Vault because the new panel ignores those preferences.

4. The current cycle - what differs in the Vault and how the shortlist accounts for it

Difference (debrief)	Evidence in the Vault	How the shortlist accounts for it
Public code contributions (new field)	5,552 rows tracked; ordinary counts decay to a cap of 60; a separate uniform population of 144 rows has 90-408 contributions and is "good not great" (mean model score 56 vs 51.5, assessment 62 vs 56).	Bonus +35 for >=90 (116 of 127 eligible are in the list; the rest have weak profiles), plus a moderate ramp 0 (<=12) to +14 (>=30) for ordinary counts. Missing/"not tracked" = no credit.

Difference (debrief)	Evidence in the Vault	How the shortlist accounts for it
Title inflation goes to the bin	Leadership titles (Manager/Director/Head/VP/Principal) held 2-5 years after graduation; none in the Archive before 7-14 years. These CVs look shiny (23% have model score >= 72 vs 4.5%).	Excluded (rules in section 5).
Cannot join within 2 months	208 rows with 90/120/150/180 days or 3-6 months (Archive max was 60). Also shiny.	Excluded (notice > 60 days).
Best people from colleges never hired from	90 colleges (380 rows) never occur in Archive or Ledger. Assessment >= 85 is 3-4x over-represented among their genuine rows (16% vs 4.5%).	Bonus up to +20 that grows from assessment 78 to 85 and requires role fit (full amount at 5+ of 6 core skills); no college penalty. 54 of 54 toppers are in the list.
Old preferences ignored	Present in the Archive/Ledger only.	College tier, city, channel, degree, employer type/size and CV gap are neutralised by partial dependence (section 6). Age, graduation year and time-since-degree are not used.
Old-boys network survives	After controlling for everything, Maharshi Dayanand Sarswati Univ. (Ajmer), Univ. of Calicut and Utkal Univ. carry a tier-1-sized effect (+5.5 to +7.4, z >= 2.5) although they are not big names.	+5 points for those three (chosen by rule in code; prestige-named colleges are never counted as network).
More fakes/repeats than before	4.4% fabricated (Archive 3.1%); 244 repeated persons.	Removed (section 5).

5. Profiles excluded and why

- **Fabricated (440)**: assessment >= 85 with two stacked positive notes, or impossible timeline (experience > years since degree + 1; age at degree < 18; age - experience < 18). Rule learnt on the Archive (mean score 9.1); no genuine-looking Archive row breaks the timeline test.
- **Title inflation (129 further)**: current title level (Senior/Lead/Principal/Manager/Director/Head/VP) reached with less time since degree (or less experience) than the minimum ever seen in the Archive for that level, or first leadership title after under 3 years of career-path time.
- **Cannot join in time (206 further)**: notice period longer than 60 days.
- **Repeated persons (221 further)**: rows are merged if they share the last 10 phone digits or a normalised e-mail (lower-case, dots and "+tag" removed). This found 244 pairs in the Vault (488 in the Archive, where twins carry identical scores - a validation of the rule). One profile per person is kept (the most complete; ties by id). Name-only matching was not used (1,487 Archive rows share a name with another person). The same rule collapses twins in the Archive and Ledger.

6. Model and selection procedure

- **Training set**: 18,904 genuine, unique Archive rows. **Model**: LightGBM regressor, 650 trees, 15 leaves, learning rate 0.03, subsample 0.8, colsample 0.7, averaged over 3 seeds. **Features**: assessment (+missing flag), aptitude, rating, KPI, awards, role-fit core-skill count, skill count, note signals (positive/negative/on-call/etc.), experience, CTC, seniority level, employers, overtime, certifications, trainings, enrolment, role, job-change gap; plus the seven preference features (college, city, channel, degree, employer type, employer size, CV gap) so the model can absorb their bias.
- **Neutralisation**: each Vault candidate is scored 24 times with the preference features replaced by draws from the Archive distribution and the outputs are averaged, i.e. the expected score if college, city, channel, degree, employer and gap were the same for everyone.
- **Final score** = neutral model score + old-boys-network bonus + code-contribution bonus + new-college-topper bonus. Excluded rows are removed, the rest sorted by score (ties by candidate_id); the top 500 become ranks 1-500. Rank 1 therefore combines a strong genuine profile with a fast-track signal.
- **Validation loop**: Ledger check of the full model (112/150), Archive CV (R^2 0.42), Archive twins have identical scores, and check_format.py returns OK.

7. What did not work

- Treating high assessment as monotonically good: fabricated rows sit at the top of the assessment range and would fill the shortlist.
- Raw 197 skill spellings/56 skill dummies and raw skill count: worse than the role-fit core-skill count (CV R^2 0.408 vs 0.424) because only role-relevant skills matter.
- Dropping the preference features instead of neutralising: kept the bias inside the remaining coefficients (e.g. experience and CTC shifted); the partial-dependence route left the true-driver coefficients unchanged.
- Career-path-sum vs total-experience mismatch as a fabrication flag: only ~-4 points, and every impossible-timeline row was already caught by the stronger rule. Name-only duplicate detection: too many chance collisions. Hard-fast-tracking every count of 24+: many natural counts were unrelated to quality, so those get only a moderate ramp.

8. External resources and AI tools (mandatory)

- Libraries: Python 3.11, pandas, numpy, LightGBM (all from Appendix A). scikit-learn was used only during exploration (cross-validation). No external datasets, no pretrained models, no downloaded data; nothing beyond the four provided CSVs is read. One constant is hard-coded: USD-INR = 83 (only to parse a handful of "$" salary strings).
- AI assistant: **Claude (Anthropic)** was used for exploratory data analysis, writing/refactoring the code in `code/` and drafting this document. All thresholds and rules were checked against the data by the team, and the team can explain every step.

9. How to run

Put train.csv, dev.csv, dev_winners.csv and test.csv in the working directory, then run `python code/main.py`. It writes submission.csv there and prints validation numbers and exclusion counts. Runtime about 40 seconds on 2 CPU cores, well under 1 GB RAM. Seeds: SEED = 7 (numpy, pandas sampling, LightGBM seeds 7, 8, 9); output is byte-identical across runs. Files: main.py, heist_clean.py (parsers), dedup.py (repeated persons).